

EXT 2: Mechanics (Ext2), M1 Mechanics (Ext2)

Projectile Motion

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Exam Equivalent Time: 97.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

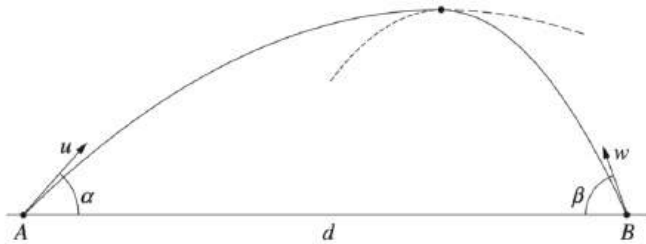
Questions

1. Mechanics, EXT2* M1 2013 HSC 13c

Points **A** and **B** are located **d** metres apart on a horizontal plane. A projectile is fired from **A** towards **B** with initial velocity **u** m/s at angle **α** to the horizontal.

At the same time, another projectile is fired from **B** towards **A** with initial velocity **w** m/s at angle **β** to the horizontal, as shown on the diagram.

The projectiles collide when they both reach their maximum height.



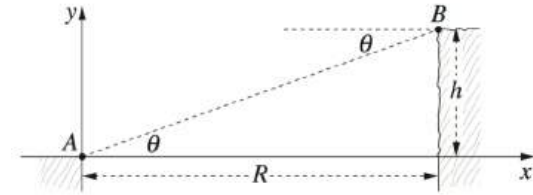
The equations of motion of a projectile fired from the origin with initial velocity **V** m/s at angle **θ** to the horizontal are

$$x = Vt \cos \theta \quad \text{and} \quad y = Vt \sin \theta - \frac{g}{2}t^2. \quad (\text{DO NOT prove this.})$$

- How long does the projectile fired from **A** take to reach its maximum height? (2 marks)
- Show that $u \sin \alpha = w \sin \beta$. (1 mark)
- Show that $d = \frac{uw}{g} \sin(\alpha + \beta)$. (2 marks)

2. Mechanics, EXT2* M1 2009 HSC 6a

Two points, **A** and **B**, are on cliff tops on either side of a deep valley. Let **h** and **R** be the vertical and horizontal distances between **A** and **B** as shown in the diagram. The angle of elevation of **B** from **A** is **θ**, so that $\theta = \tan^{-1}\left(\frac{h}{R}\right)$.



At time **t = 0**, projectiles are fired simultaneously from **A** and **B**. The projectile from **A** is aimed at **B**, and has initial speed **U** at an angle of **θ** above the horizontal. The projectile from **B** is aimed at **A** and has initial speed **V** at an angle **θ** below the horizontal.

The equations of motion for the projectile from **A** are

$$x_1 = Ut \cos \theta \quad \text{and} \quad y_1 = Ut \sin \theta - \frac{1}{2}gt^2,$$

and the equations for the motion of the projectile from **B** are

$$x_2 = R - Vt \cos \theta \quad \text{and} \quad y_2 = h - Vt \sin \theta - \frac{1}{2}gt^2, \quad (\text{DO NOT prove these equations.})$$

- Let **T** be the time at which $x_1 = x_2$.

Show that $T = \frac{R}{(U + V) \cos \theta}$. (1 mark)

- Show that the projectiles collide. (2 marks)

- If the projectiles collide on the line $x = \lambda R$, where $0 < \lambda < 1$, show that

$$V = \left(\frac{1}{\lambda} - 1\right)U. \quad (1 \text{ mark})$$

3. Mechanics, EXT2* M1 2007 HSC 7b

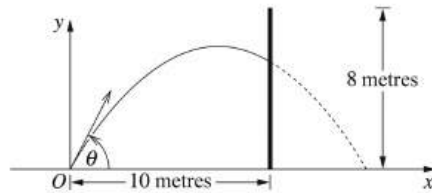
A small paintball is fired from the origin with initial velocity **14** metres per second towards an eight-metre high barrier. The origin is at ground level, **10** metres from the base of the barrier.

The equations of motion are

$$x = 14t \cos \theta$$

$$y = 14t \sin \theta - 4.9t^2$$

where θ is the angle to the horizontal at which the paintball is fired and t is the time in seconds. (Do NOT prove these equations of motion)



i. Show that the equation of trajectory of the paintball is

$$y = mx - \left(\frac{1 + m^2}{40} \right) x^2, \text{ where } m = \tan \theta. \text{ (2 mark)}$$

ii. Show that the paintball hits the barrier at height h metres when

$$m = 2 \pm \sqrt{3 - 0.4h}$$

Hence determine the maximum value of h . (2 marks)

iii. There is a large hole in the barrier. The bottom of the hole is **3.9** metres above the ground and the top of the hole is **5.9** metres above the ground. The paintball passes through the hole if m is in one of two intervals. One interval is $2.8 \leq m \leq 3.2$.

Find the other interval. (2 marks)

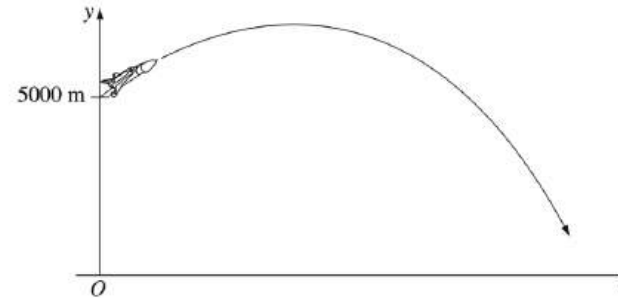
iv. Show that, if the paintball passes through the hole, the range is

$$\frac{40m}{1 + m^2} \text{ metres.}$$

Hence find the widths of the two intervals in which the paintball can land at ground level on the other side of the barrier. (3 marks)

4. Mechanics, EXT2* M1 2005 6b

An experimental rocket is at a height of 5000 m, ascending with a velocity of $200\sqrt{2} \text{ m s}^{-1}$ at an angle of 45° to the horizontal, when its engine stops.



After this time, the equations of motion of the rocket are:

$$x = 200t$$

$$y = -4.9t^2 + 200t + 5000,$$

where t is measured in seconds after the engine stops. (Do NOT show this.)

i. What is the maximum height the rocket will reach, and when will it reach this height? (2 marks)

ii. The pilot can only operate the ejection seat when the rocket is descending at an angle between 45° and 60° to the horizontal. What are the earliest and latest times that the pilot can operate the ejection seat? (3 marks)

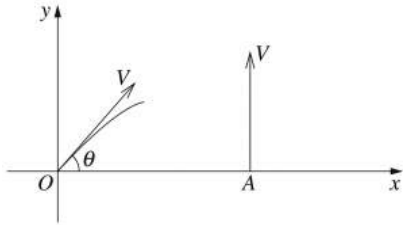
iii. For the parachute to open safely, the pilot must eject when the speed of the rocket is no more than 350 m s^{-1} . What is the latest time at which the pilot can eject safely? (2 marks)

5. Mechanics, EXT2* M1 2006 HSC 6a

Two particles are fired simultaneously from the ground at time $t = 0$.

Particle 1 is projected from the origin at an angle θ , $0 < \theta < \frac{\pi}{2}$, with an initial velocity V .

Particle 2 is projected vertically upward from the point A , at a distance a to the right of the origin, also with an initial velocity of V .



It can be shown that while both particles are in flight, Particle 1 has equations of motion:

$$x = Vt \cos \theta$$

$$y = Vt \sin \theta - \frac{1}{2}gt^2,$$

and Particle 2 has equations of motion:

$$x = a$$

$$y = Vt - \frac{1}{2}gt^2. \quad \text{Do NOT prove these equations of motion.}$$

Let L be the distance between the particles at time t .

i. Show that, while both particles are in flight,

$$L^2 = 2V^2t^2(1 - \sin \theta) - 2aVt \cos \theta + a^2. \quad (2 \text{ marks})$$

ii. An observer notices that the distance between the particles in flight first decreases, then increases.

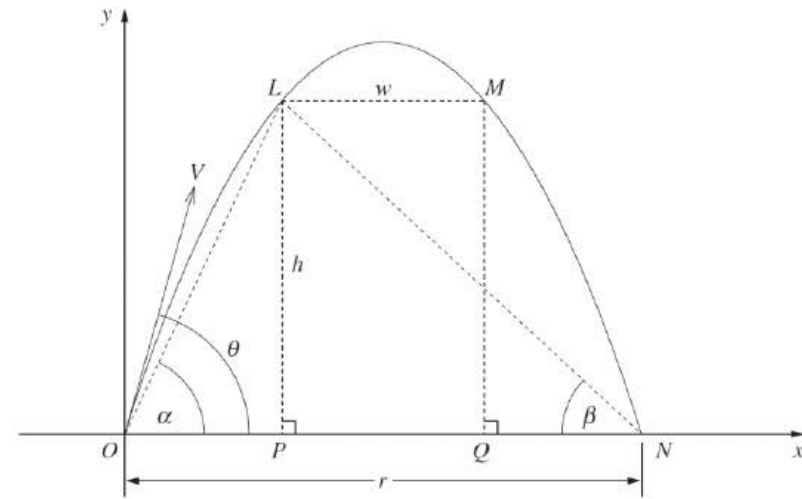
Show that the distance between the particles in flight is smallest when

$$t = \frac{a \cos \theta}{2V(1 - \sin \theta)} \quad \text{and that this smallest distance is } a\sqrt{\frac{1 - \sin \theta}{2}}. \quad (3 \text{ marks})$$

iii. Show that the smallest distance between the two particles in flight occurs while Particle 1 is ascending if

$$V > \sqrt{\frac{ag \cos \theta}{2 \sin \theta (1 - \sin \theta)}}. \quad (1 \text{ mark})$$

6. Mechanics, EXT2* M1 2008 HSC 7



A projectile is fired from O with velocity V at an angle of inclination θ across level ground. The projectile passes through the points L and M , which are both h metres above the ground, at times t_1 and t_2 respectively. The projectile returns to the ground at N .

The equations of motion of the projectile are

$$x = Vt \cos \theta \quad \text{and} \quad y = Vt \sin \theta - \frac{1}{2}gt^2. \quad (\text{Do NOT prove this.})$$

i. Show that $t_1 + t_2 = \frac{2V}{g} \sin \theta$ AND $t_1 t_2 = \frac{2h}{g}$. (2 marks)

Let $\angle LON = \alpha$ and $\angle LNO = \beta$. It can be shown that

$$\tan \alpha = \frac{h}{Vt_1 \cos \theta} \quad \text{and} \quad \tan \beta = \frac{h}{Vt_2 \cos \theta}. \quad (\text{Do NOT prove this.})$$

ii. Show that $\tan \alpha + \tan \beta = \tan \theta$. (2 marks)

iii. Show that $\tan \alpha \tan \beta = \frac{gh}{2V^2 \cos^2 \theta}$. (1 mark)

Let $ON = r$ and $LM = w$.

iv. Show that $r = h(\cot \alpha + \cot \beta)$ and $w = h(\cot \beta - \cot \alpha)$. (2 marks)

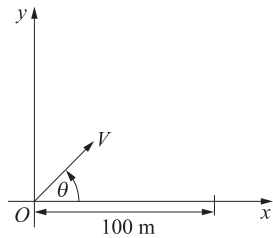
Let the gradient of the parabola at L be $\tan \phi$.

v. Show that $\tan \phi = \tan \alpha - \tan \beta$. (3 marks)

vi. Show that $\frac{w}{\tan \phi} = \frac{r}{\tan \theta}$. (2 marks)

7. Mechanics, EXT2* M1 2017 HSC 13c

A golfer hits a golf ball with initial speed $V \text{ ms}^{-1}$ at an angle θ to the horizontal. The golf ball is hit from one side of a lake and must have a horizontal range of 100 m or more to avoid landing in the lake.



Neglecting the effects of air resistance, the equations describing the motion of the ball are

$$x = Vt \cos \theta$$

$$y = Vt \sin \theta - \frac{1}{2}gt^2,$$

where t is the time in seconds after the ball is hit and g is the acceleration due to gravity in ms^{-2} . Do NOT prove these equations.

i. Show that the horizontal range of the golf ball is

$$\frac{V^2 \sin 2\theta}{g} \text{ metres. (2 marks)}$$

ii. Show that if $V^2 < 100g$ then the horizontal range of the ball is less than 100 m. (1 mark)

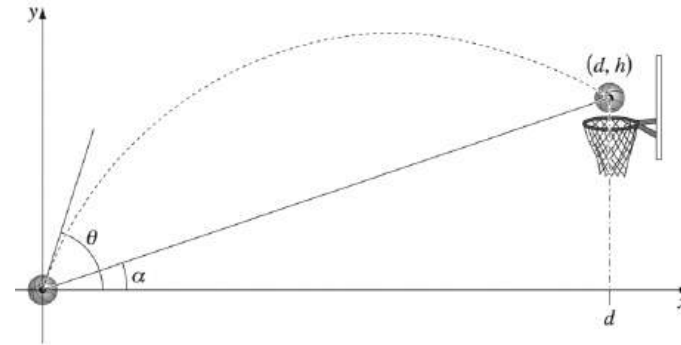
It is now given that $V^2 = 200g$ and that the horizontal range of the ball is 100 m or more.

iii. Show that $\frac{\pi}{12} \leq \theta \leq \frac{5\pi}{12}$. (2 marks)

iv. Find the greatest height the ball can achieve. (2 marks)

8. Mechanics, EXT2* M1 2010 HSC 6b

A basketball player throws a ball with an initial velocity $v \text{ m/s}$ at an angle of θ to the horizontal. At the time the ball is released its centre is at $(0, 0)$, and the player is aiming for the point (d, h) as shown on the diagram. The line joining $(0, 0)$ and (d, h) makes an angle α with the horizontal, where $0 < \alpha < \frac{\pi}{2}$.



Assume that at time t seconds after the ball is thrown its centre is at the point (x, y) , where

$$x = vt \cos \theta$$

$$y = vt \sin \theta - 5t^2. \quad (\text{DO NOT prove this.})$$

i. If the centre of the ball passes through (d, h) show that

$$v^2 = \frac{5d}{\cos \theta \sin \theta - \cos^2 \theta \tan \alpha} \quad (3 \text{ marks})$$

ii. (1) What happens to v as $\theta \rightarrow \alpha$? (1 mark)

(2) What happens to v as $\theta \rightarrow \frac{\pi}{2}$? (1 mark)

iii. For a fixed value of α , let $F(\theta) = \cos \theta \sin \theta - \cos^2 \theta \tan \alpha$.

Show that $F'(\theta) = 0$ when $\tan 2\theta \tan \alpha = -1$ (2 marks)

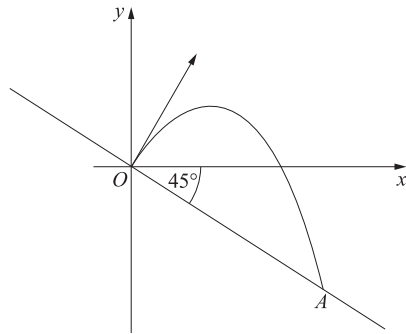
iv. Using part (a)(ii)* or otherwise show that $F'(\theta) = 0$, when $\theta = \frac{\alpha}{2} + \frac{\pi}{4}$. (1 mark)

*Please note for the purposes of this question, part (a)(ii) showed that when $\tan A \tan B = -1$, then $A - B = \frac{\pi}{2}$

v. Explain why v^2 is a minimum when $\theta = \frac{\alpha}{2} + \frac{\pi}{4}$ (2 marks)

9. Mechanics, EXT2* M1 2019 HSC 13d

The point O is on a sloping plane that forms an angle of 45° to the horizontal. A particle is projected from the point O . The particle hits a point A on the sloping plane as shown in the diagram.



The equation of the line OA is $y = -x$. The equations of motion of the particle are

$$x = 18t$$

$$y = 18\sqrt{3}t - 5t^2,$$

where t is the time in seconds after projection. Do NOT prove these equations.

- Find the distance OA between the point of projection and the point where the particle hits the sloping plane. (2 marks)
- What is the size of the acute angle that the path of the particle makes with the sloping plane as the particle hits the point A ? (3 marks)

Worked Solutions

1. Mechanics, EXT2* M1 2013 HSC 13c

$$i. \quad y = Vt \sin \theta - \frac{g}{2}t^2$$

$$\text{Projectile A} \Rightarrow V = u, \quad \theta = \alpha$$

$$y = ut \sin \alpha - \frac{g}{2}t^2$$

$$\dot{y} = u \sin \alpha - gt$$

Max height when $\dot{y} = 0$

$$0 = u \sin \alpha - gt$$

$$gt = u \sin \alpha$$

$$t = \frac{u}{g} \sin \alpha$$

$\therefore$ Projectile from A reaches max height at

$$t = \frac{u}{g} \sin \alpha \text{ seconds}$$

ii. Show that $u \sin \alpha = w \sin \beta$

$$\text{Projectile B} \Rightarrow V = w, \quad \theta = \beta$$

$$y = wt \sin \beta - \frac{g}{2}t^2$$

$$\dot{y} = w \sin \beta - gt$$

Max height when $\dot{y} = 0$

$$t = \frac{w}{g} \sin \beta$$

Projectiles collide at max heights

Since they were fired at the same time

$$\frac{u}{g} \sin \alpha = \frac{w}{g} \sin \beta$$

$$\therefore u \sin \alpha = w \sin \beta \text{ ... as required}$$

iii Show $d = \frac{uw}{g} \sin(\alpha + \beta)$:

IMPORTANT: Part (i) in this example leads students to a very quick solution. Always look to previous parts for clues to direct your strategy.

Find x -values for each projectile at max height

Projectile A

$$\begin{aligned}x_1 &= ut \cos \alpha \\&= u \left(\frac{u}{g} \sin \alpha \right) \cos \alpha \\&= \frac{u^2}{g} \sin \alpha \cos \alpha\end{aligned}$$

Projectile B

$$\begin{aligned}x_2 &= wt \cos \beta \\&= w \left(\frac{w}{g} \sin \beta \right) \cos \beta \\&= \frac{w^2}{g} \sin \beta \cos \beta\end{aligned}$$

$$\begin{aligned}d &= x_1 + x_2 \\&= \frac{u^2}{g} \sin \alpha \cos \alpha + \frac{w^2}{g} \sin \beta \cos \beta \\&= \frac{u}{g} (u \sin \alpha) \cos \alpha + \frac{w}{g} (w \sin \beta) \cos \beta \\&= \frac{u}{g} (w \sin \beta) \cos \alpha + \frac{w}{g} (u \sin \alpha) \cos \beta \quad (\text{part (ii)}) \\&= \frac{uw}{g} (\sin \alpha \cos \beta + \cos \alpha \sin \beta) \\&= \frac{uw}{g} \sin(\alpha + \beta) \quad \dots \text{as required}\end{aligned}$$

♦ Mean mark 39%
IMPORTANT: Students should direct their calculations by reverse engineering the required result. $\sin(\alpha + \beta)$ in the proof means that $\sin \alpha \cos \beta + \cos \alpha \sin \beta$ will appear in the working calculations.

2. Mechanics, EXT2* M1 2009 HSC 6a

i. Let $x_1 = x_2$ at $t = T$

$$\begin{aligned}UT \cos \theta &= R - VT \cos \theta \\UT \cos \theta + VT \cos \theta &= R \\T \cos \theta (U + V) &= R \\\therefore T &= \frac{R}{(U + V) \cos \theta} \quad \dots \text{as required}\end{aligned}$$

ii. If particles collide, $y_1 = y_2$ at $t = T$

$$\begin{aligned}y_1 &= UT \sin \theta - \frac{1}{2} g T^2 \\&= \frac{UR \sin \theta}{(U + V) \cos \theta} - \frac{1}{2} g T^2\end{aligned}$$

$$\begin{aligned}y_2 &= h - VT \sin \theta - \frac{1}{2} g T^2 \\&= R \tan \theta - \frac{VR \sin \theta}{(U + V) \cos \theta} - \frac{1}{2} g T^2 \\&= R \left(\frac{\left(\frac{\sin \theta}{\cos \theta} \right) (U + V) \cos \theta - V \sin \theta}{(U + V) \cos \theta} \right) - \frac{1}{2} g T^2 \\&= \frac{R}{(U + V) \cos \theta} (U \sin \theta + V \sin \theta - V \sin \theta) - \frac{1}{2} g T^2 \\&= \frac{UR \sin \theta}{(U + V) \cos \theta} - \frac{1}{2} g T^2 \\&= y_1\end{aligned}$$

$\therefore$ Particles collide since $y_1 = y_2$ at $t = T$.

iii. Particles collide on line $x = \lambda R$, i.e. where $t = T$

$$\begin{aligned}\lambda R &= UT \cos \theta \\&= U \frac{R}{(U + V) \cos \theta} \cos \theta\end{aligned}$$

♦♦♦ Mean mark data not available although "few" students were able to complete this part.

MARKER'S COMMENT: Better responses showed $y_1 = y_2$, or $y_1 - y_2 = 0$ which eliminated $-\frac{1}{2}gt^2$ from the algebra. Substituting $\tan \theta = h/R$ was a key element in completing this proof.

MARKER'S COMMENT: Students must show sufficient working in proofs to convince markers they have derived the answer themselves.

$$= \frac{UR}{(U+V)}$$

$$(U+V)(\lambda R) = UR$$

$$U+V = \frac{UR}{\lambda R}$$

$$V = \frac{U}{\lambda} - U = \left(\frac{1}{\lambda} - 1\right)U \dots \text{as required}$$

3. Mechanics, EXT2* M1 2007 HSC 7b

i. $x = 14t \cos \theta \dots (1)$

$$y = 14t \sin \theta - 4.9t^2 \dots (2)$$

Substitute $t = \frac{x}{14 \cos \theta}$ from (1) into (2)

$$y = 14 \left(\frac{x}{14 \cos \theta} \right) \sin \theta - 4.9 \left(\frac{x}{14 \cos \theta} \right)^2$$

$$= x \tan \theta - \left(\frac{4.9}{14^2} \right) \left(\frac{x^2}{\cos^2 \theta} \right)$$

$$= x \tan \theta - \frac{x^2}{40} \sec^2 \theta = x \tan \theta - \frac{x^2}{40} (1 + \tan^2 \theta)$$

$$= mx - \left(\frac{1+m^2}{40} \right) x^2 \quad (\text{Given } m = \tan \theta)$$

ii. Show paintball hits at h when

$$m = 2 \pm \sqrt{3 - 0.4h}$$

i.e. $y = h$ when $x = 10$

$$10m - \left(\frac{1+m^2}{40} \right) \cdot 10^2 = h$$

$$10m - \frac{5}{2}(1+m^2) = h$$

$$20m - 5 - 5m^2 = 2h$$

$$5m^2 - 20m + 2h + 5 = 0$$

Using the quadratic formula

$$m = \frac{20 \pm \sqrt{(-20)^2 - 4 \cdot 5 \cdot (2h+5)}}{2 \cdot 5}$$

$$= \frac{20 \pm \sqrt{400 - 40h - 100}}{10} = \frac{20 \pm \sqrt{300 - 40h}}{10}$$

$$= \frac{20 \pm 10\sqrt{3 - 0.4h}}{10}$$

$$= 2 \pm \sqrt{3 - 0.4h} \dots \text{as required}$$

Find maximum h

$$\sqrt{3 - 0.4h} \geq 0$$

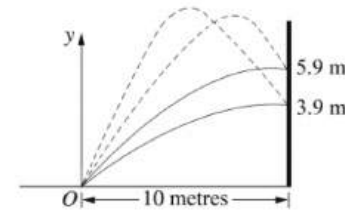
$$3 - 0.4h \geq 0$$

$$0.4h \leq 3$$

$$h \leq 7.5$$

$\therefore$ Maximum $h = 7.5$ m

iii.



Using part (ii)

When $h = 3.9$

$$m = 2 \pm \sqrt{3 - 0.4(3.9)}$$

$$= 2 \pm \sqrt{1.44}$$

$$= 2 \pm 1.2$$

$$= 3.2 \text{ or } 0.8$$

When $h = 5.9$

$$m = 2 \pm \sqrt{3 - 0.4(5.9)}$$

$$= 2 \pm \sqrt{0.64}$$

$$= 2 \pm 0.8$$

$$= 2.8 \text{ or } 1.2$$

$\therefore$ The other interval is $0.8 \leq m \leq 1.2$

iv. Find x when $y = 0$

$$mx - \left(\frac{1+m^2}{40}\right)x^2 = 0$$

$$x \left[m - \left(\frac{1+m^2}{40}\right)x \right] = 0$$

$$\left(\frac{1+m^2}{40}\right)x = m, \quad x \neq 0$$

$$\therefore x = \frac{40m}{1+m^2} \dots \text{as required}$$

Consider the interval $2.8 \leq m \leq 3.2$

When $m = 2.8$

$$\Rightarrow x = \frac{40(2.8)}{1+2.8^2} = 12.669\dots \text{ m}$$

When $m = 3.2$

$$\Rightarrow x = \frac{40(3.2)}{1+3.2^2} = 11.387\dots \text{ m}$$

$\therefore$ Landing width interval

$$= 12.669\dots - 11.387\dots$$

$$= 1.281\dots$$

$$= 1.3 \text{ m to 1 d.p.}$$

Consider the interval $0.8 \leq m \leq 1.2$

When $m = 0.8$

$$\Rightarrow x = \frac{40(0.8)}{1+0.8^2} = 19.512\dots \text{ m}$$

When $m = 1.2$

$$\Rightarrow x = \frac{40(1.2)}{1+1.2^2} = 19.672\dots$$

Since interval includes $m = 1$ where the paintball has maximum range.

$$x_{\max} = \frac{40(1)}{1+1^2} = 20 \text{ m}$$

$\therefore$ Landing width interval

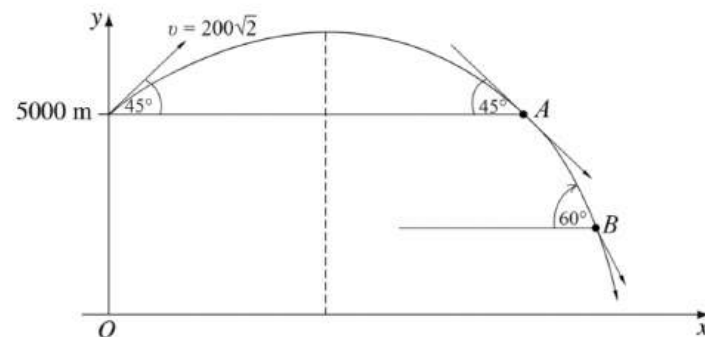
$$= 20 - 19.512\dots$$

$$= 0.487\dots$$

$$= 0.5 \text{ m (to 1 d.p.)}$$

4. Mechanics, EXT2* M1 2005 6b

i.



$$y = -4.9t^2 + 200t + 5000$$

$$\dot{y} = -9.8t + 200$$

Max height occurs when $\dot{y} = 0$

$$9.8t = 200$$

$$t = 20.408\dots$$

$$= 20.4 \text{ seconds (to 1 d.p.)}$$

When $t = 20.408\dots$

$$y = -4.9(20.408\dots)^2 + 200(20.408\dots) + 5000$$

$$= 7040.816\dots$$

$$= 7041 \text{ m (to nearest metre)}$$

$\therefore$ The rocket will reach a maximum height of

7041 metres when $t = 20.4$ seconds.

ii. The rocket is descending at 45° at point A on the graph.

The symmetry of the parabolic motion means that A occurs when $t = 2 \times$ time to reach max height, or 40.8 seconds.

Point B occurs when the rocket is descending at 60° to the horizontal.

At B,

$$-\tan 60^\circ = \frac{\dot{y}}{\dot{x}}$$

(The gradient of the projectile becomes negative after reaching its max height.)

$$\dot{y} = -9.8t + 200$$

$$\dot{x} = \frac{d}{dt}(200t) = 200$$

$$\therefore -\sqrt{3} = \frac{-9.8t + 200}{200}$$

$$-200\sqrt{3} = -9.8t + 200$$

$$9.8t = 200 + 200\sqrt{3}$$

$$t = \frac{200 + 200\sqrt{3}}{9.8}$$

$$= \frac{546.410\dots}{9.8}$$

$$= 55.756\dots$$

$$= 55.8 \text{ seconds (nearest second)}$$

$\therefore$ The pilot can operate the ejection seat between $t = 40.8$ and $t = 55.8$ seconds.

$$\text{iii. } v^2 = (\dot{x})^2 + (\dot{y})^2$$

When $v = 350$

$$350^2 = 200^2 + (-9.8t + 200)^2$$

$$122\,500 = 40\,000 + (-9.8t + 200)^2$$

$$(-9.8t + 200)^2 = 82\,500$$

$$-9.8t + 200 = \pm \sqrt{82\,500}$$

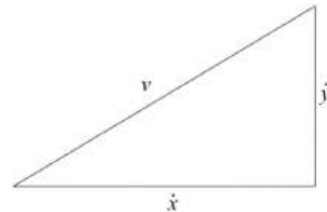
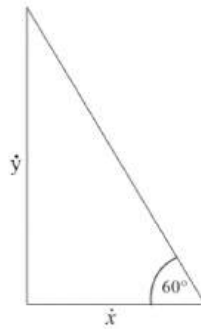
$$9.8t = 200 \pm \sqrt{82\,500}$$

$$t = \frac{200 \pm \sqrt{82\,500}}{9.8}$$

$$= 49.717\dots \quad (t > 0)$$

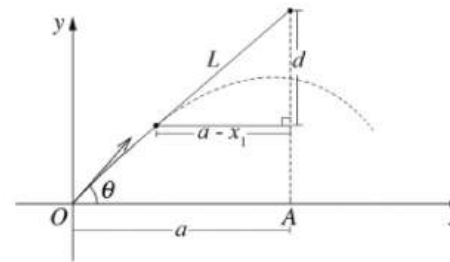
$$= 49.7 \text{ seconds (to 1 d.p.)}$$

$\therefore$ The latest time the pilot can eject safely is when $t = 49.7$ seconds.



5. Mechanics, EXT2* M1 2006 HSC 6a

i.



$$\text{Show } L^2 = 2V^2t^2(1 - \sin \theta) - 2aVt \cos \theta + a^2$$

Consider P_1

$$x_1 = Vt \cos \theta$$

$$y_1 = Vt \sin \theta - \frac{1}{2}gt^2$$

Consider P_2

$$x_2 = a$$

$$y_2 = Vt - \frac{1}{2}gt^2$$

Let d = Vertical distance between particles

$$d = Vt - \frac{1}{2}gt^2 - \left(Vt \sin \theta - \frac{1}{2}gt^2 \right)$$

$$d = Vt(1 - \sin \theta)$$

Using Pythagoras:

$$L^2 = (a - x_1)^2 + d^2$$

$$= (a - Vt \cos \theta)^2 + V^2t^2(1 - \sin \theta)^2$$

$$= a^2 - 2aVt \cos \theta + V^2t^2 \cos^2 \theta + V^2t^2(1 - 2\sin \theta + \sin^2 \theta)$$

$$= a^2 - 2aVt \cos \theta + V^2t^2(\cos^2 \theta + \sin^2 \theta + 1 - 2\sin \theta)$$

$$= a^2 - 2aVt \cos \theta + V^2t^2(2 - 2\sin \theta)$$

$$= 2V^2t^2(1 - \sin \theta) - 2aVt \cos \theta + a^2 \dots \text{ as required.}$$

$$\text{ii. } L^2 = 2V^2t^2(1 - \sin \theta) - 2aVt \cos \theta + a^2$$

$$\frac{d(L^2)}{dt} = 4V^2t(1 - \sin \theta) - 2aV \cos \theta$$

Max or min when $\frac{d(L^2)}{dt} = 0$

$$4V^2 t (1 - \sin \theta) = 2aV \cos \theta$$

$$t = \frac{2aV \cos \theta}{4V^2(1 - \sin \theta)}$$

$$= \frac{a \cos \theta}{2V(1 - \sin \theta)}$$

$$\frac{d^2(L^2)}{dt^2} = 4V^2(1 - \sin \theta) > 0 \text{ for } V > 0, 0 < \theta < \frac{\pi}{2}$$

$\therefore L^2$ is a minimum

$$\therefore L \text{ is a minimum when } t = \frac{a \cos \theta}{2V(1 - \sin \theta)}$$

Show minimum distance is $a\sqrt{\frac{1 - \sin \theta}{2}}$

When $t = \frac{a \cos \theta}{2V(1 - \sin \theta)}$

$$L^2 = 2V^2 \left(\frac{a^2 \cos^2 \theta}{4V^2(1 - \sin \theta)^2} \right) (1 - \sin \theta)$$

$$- 2aV \left(\frac{a \cos \theta}{2V(1 - \sin \theta)} \right) \cos \theta + a^2$$

$$= \frac{a^2 \cos^2 \theta}{2(1 - \sin \theta)} - \frac{a^2 \cos^2 \theta}{(1 - \sin \theta)} + a^2$$

$$= a^2 \left[\frac{\cos^2 \theta - 2 \cos^2 \theta + 2(1 - \sin \theta)}{2(1 - \sin \theta)} \right]$$

$$= a^2 \left[\frac{-\cos^2 \theta + 2 - 2 \sin \theta}{2(1 - \sin \theta)} \right]$$

$$= a^2 \left[\frac{-(1 - \sin^2 \theta) + 2 - 2 \sin \theta}{2(1 - \sin \theta)} \right]$$

$$= a^2 \left[\frac{\sin^2 \theta - 2 \sin \theta + 1}{2(1 - \sin \theta)} \right]$$

$$= a^2 \left[\frac{(1 - \sin \theta)^2}{2(1 - \sin \theta)} \right]$$

$$= a^2 \left[\frac{(1 - \sin \theta)}{2} \right]$$

$$\therefore L = \sqrt{\frac{a^2(1 - \sin \theta)}{2}}$$

$$= a\sqrt{\frac{1 - \sin \theta}{2}} \dots \text{ as required.}$$

iii. Smallest distance occurs when

$$t = \frac{a \cos \theta}{2V(1 - \sin \theta)}$$

If P_1 is ascending, $\dot{y}_1 > 0$

$$y_1 = Vt \sin \theta - \frac{1}{2}gt^2$$

$$\dot{y}_1 = V \sin \theta - gt$$

$$\therefore V \sin \theta - g \left(\frac{a \cos \theta}{2V(1 - \sin \theta)} \right) > 0$$

$$2V^2 \sin \theta (1 - \sin \theta) - ag \cos \theta > 0$$

$$2V^2 \sin \theta (1 - \sin \theta) > ag \cos \theta$$

$$V^2 > \frac{ag \cos \theta}{2 \sin \theta (1 - \sin \theta)}$$

$$V > \sqrt{\frac{ag \cos \theta}{2 \sin \theta (1 - \sin \theta)}} \dots \text{ as required.}$$

6. Mechanics, EXT2* M1 2008 HSC 7

i. At $y = h$,

$$-\frac{1}{2}gt^2 + Vt \sin \theta = h$$

$$gt^2 - 2Vt \sin \theta + 2h = 0$$

Roots are t_1 and t_2

$$\therefore t_1 + t_2 = -\frac{b}{a}$$

$$= \frac{2V}{g} \sin \theta \dots \text{as required}$$

$$\therefore t_1 t_2 = \frac{c}{a}$$

$$= \frac{2h}{g} \dots \text{as required}$$

ii. Show $\tan \alpha + \tan \beta = \tan \theta$

$$\begin{aligned} \text{LHS} &= \frac{h}{Vt_1 \cos \theta} + \frac{h}{Vt_2 \cos \theta} \\ &= \frac{h(t_1 + t_2)}{t_1 t_2 V \cos \theta} \\ &= \frac{h\left(\frac{2V}{g} \sin \theta\right)}{\left(\frac{2h}{g}\right) V \cos \theta}, \quad (\text{using part (i)}) \\ &= \frac{\frac{2Vh}{g}}{\frac{2Vh}{g}} \times \frac{\sin \theta}{\cos \theta} \\ &= \tan \theta \dots \text{as required.} \end{aligned}$$

iii. Show $\tan \alpha \tan \beta = \frac{gh}{2V^2 \cos^2 \theta}$

$$\begin{aligned} \text{LHS} &= \frac{h}{Vt_1 \cos \theta} \times \frac{h}{Vt_2 \cos \theta} \\ &= \frac{h^2}{t_1 t_2 V^2 \cos^2 \theta} \end{aligned}$$

$$\begin{aligned} &= \frac{h^2}{\frac{2h}{g} \cdot V^2 \cos^2 \theta} \\ &= \frac{gh}{2V^2 \cos^2 \theta} \dots \text{as required.} \end{aligned}$$

iv. From the diagram,

$$r = ON = OP + PN$$

In $\triangle OLP$,

$$\tan \alpha = \frac{h}{OP}$$

$$OP = h \cot \alpha$$

In $\triangle NLP$,

$$\tan \beta = \frac{h}{PN}$$

$$PN = h \cot \beta$$

$$\begin{aligned} \therefore r &= h \cot \alpha + h \cot \beta \\ &= h(\cot \alpha + \cot \beta) \dots \text{as required.} \end{aligned}$$

$$\begin{aligned} w &= r - (OP + QN) \\ &= r - 2 \times OP, \quad (\text{symmetry of parabola}) \\ &= h \cot \alpha + h \cot \beta - 2h \cot \alpha \\ &= h(\cot \beta - \cot \alpha) \dots \text{as required.} \end{aligned}$$

v. Show $\tan \phi = \tan \alpha - \tan \beta$

At L , $t = t_1$ and

$$\tan \phi = \frac{\dot{y}}{\dot{x}}$$

$$\dot{y} = V \sin \theta - gt$$

$$\dot{x} = V \cos \theta$$

$$\begin{aligned} \therefore \tan \phi &= \frac{V \sin \theta - gt_1}{V \cos \theta} \\ &= \tan \theta - \frac{gt_1}{V \cos \theta} \\ &= \tan \theta - \frac{g}{V \cos \theta} \times \frac{h}{V \cos \theta \tan \alpha} \end{aligned}$$

$$\begin{aligned}
&= \tan \theta - \frac{gh}{V^2 \cos^2 \theta \tan \alpha} \\
&= \tan \theta - 2 \tan \beta, \quad (\text{from part (iii)}) \\
&= \tan \alpha + \tan \beta - 2 \tan \beta, \quad (\text{from part (ii)}) \\
&= \tan \alpha - \tan \beta \dots \text{as required.}
\end{aligned}$$

Show $\frac{w}{\tan \phi} = \frac{r}{\tan \theta}$

$$\begin{aligned}
\frac{w}{r} &= \frac{h(\cot \beta - \cot \alpha)}{h(\cot \alpha + \cot \beta)} \\
&= \frac{\frac{1}{\tan \beta} - \frac{1}{\tan \alpha}}{\frac{1}{\tan \alpha} + \frac{1}{\tan \beta}} \times \frac{\tan \alpha \tan \beta}{\tan \alpha \tan \beta} \\
&= \frac{\tan \alpha - \tan \beta}{\tan \beta + \tan \alpha} \\
&= \frac{\tan \phi}{\tan \theta}
\end{aligned}$$

$$\therefore \frac{w}{\tan \phi} = \frac{r}{\tan \theta} \dots \text{as required.}$$

7. Mechanics, EXT2* M1 2017 HSC 13c

i. Find t when $y = 0$:

$$\frac{1}{2}gt^2 = Vt \sin \theta$$

$$\frac{1}{2}gt = V \sin \theta$$

$$t = \frac{2V \sin \theta}{g}$$

Horizontal range (x) when $t = \frac{2V \sin \theta}{g}$:

$$\begin{aligned}
x &= V \cdot \frac{2V \sin \theta}{g} \cos \theta \\
&= \frac{V^2 2 \sin \theta \cos \theta}{g} \\
&= \frac{V^2 \sin 2\theta}{g} \dots \text{as required}
\end{aligned}$$

ii. If $V^2 < 100g$

$$x < \frac{100g \sin 2\theta}{g}$$

$$x < 100 \sin 2\theta$$

Since $-1 \leq 2\theta \leq 1$,

$$x < 100 \text{ metres}$$

iii. $V^2 = 200g$, $x \geq 100$

$$\frac{200g \cdot \sin 2\theta}{g} \geq 100$$

$$\sin 2\theta \geq \frac{1}{2}$$

$$\therefore \frac{\pi}{6} \leq 2\theta \leq \frac{5\pi}{6}$$

$$\therefore \frac{\pi}{12} \leq \theta \leq \frac{5\pi}{12} \dots \text{as required}$$

♦ Mean mark 44%.

iv. Max height occurs when

$$t = \frac{1}{2} \times \text{time of flight}$$

$$= \frac{V \sin \theta}{g}$$

Find y when $t = \frac{V \sin \theta}{g}$

$$y = V \cdot \frac{V \sin \theta}{g} \cdot \sin \theta - \frac{1}{2} g \left(\frac{V \sin \theta}{g} \right)^2$$

$$= \frac{V^2 \sin^2 \theta}{g} - \frac{1}{2} \cdot \frac{V^2 \sin^2 \theta}{g}$$

$$= \frac{V^2 \sin^2 \theta}{2g}$$

Max height when $\theta = \frac{5\pi}{12}$ (steepest angle), $V^2 = 200g$ (given)

$$y_{\max} = \frac{200g \cdot \sin^2 \left(\frac{5\pi}{12} \right)}{2g}$$

$$= 100 \sin^2 \left(\frac{5\pi}{12} \right)$$

$$= 50 \left(1 - \cos \left(\frac{5\pi}{6} \right) \right)$$

$$= 50 \left(1 + \frac{\sqrt{3}}{2} \right)$$

$$= 25(2 + \sqrt{3}) \text{ metres}$$

♦♦ Mean mark 35%.

8. Mechanics, EXT2* M1 2010 HSC 6b

i. Ball passes through (d, h) .

Find t when $x = d$:

$$d = vt \cos \theta$$

$$\therefore t = \frac{d}{v \cos \theta}$$

Since $y = h$ when $t = \frac{d}{v \cos \theta}$

$$h = vt \sin \theta - 5t^2$$

$$= v \left(\frac{d}{v \cos \theta} \right) \sin \theta - 5 \frac{d^2}{v^2 \cos^2 \theta}$$

$$\frac{5d^2}{v^2 \cos^2 \theta} = \frac{d \sin \theta}{\cos \theta} - h$$

$$\frac{5d^2}{v^2} = d \sin \theta \cos \theta - \cos^2 \theta \times h$$

$$\frac{v^2}{5d^2} = \frac{1}{d \sin \theta \cos \theta - \cos^2 \theta \times h}$$

$$v^2 = \frac{5d^2}{d \sin \theta \cos \theta - \cos^2 \theta \times h}$$

$$= \frac{5d}{\cos \theta \sin \theta - \cos^2 \theta \times \frac{h}{d}}$$

$$= \frac{5d}{\cos \theta \sin \theta - \cos^2 \theta \tan \alpha} \dots \text{as required}$$

♦ Mean mark 43%

IMPORTANT: It is critical to use the relationship $\tan \alpha = \frac{h}{d}$ to achieve the required proof.

ii. (1) As $\theta \rightarrow \alpha$

$$\cos \theta \sin \theta - \cos^2 \theta \tan \alpha \rightarrow 0$$

$$v \rightarrow \infty$$

ii. (2) As $\theta \rightarrow \frac{\pi}{2}$

$$\cos \theta \sin \theta - \cos^2 \theta \tan \theta \rightarrow 0$$

$$v \rightarrow \infty$$

iii. Show $\tan 2\theta \tan \alpha = -1$ when $F'(\theta) = 0$.

$$F(\theta) = \cos \theta \sin \theta - \cos^2 \theta \tan \alpha$$

♦ Mean marks of 37% and 32% for parts (ii)(1) and (ii)(2) respectively.

$$\begin{aligned}
 F'(\theta) &= \cos \theta \cos \theta + \sin \theta (-\sin \theta) - 2 \cos \theta (-\sin \theta) \tan \alpha \\
 &= \cos^2 \theta - \sin^2 \theta + \sin 2\theta \tan \alpha \\
 &= \cos 2\theta + \sin 2\theta \tan \alpha
 \end{aligned}$$

When $F'(\theta) = 0$

$$\cos 2\theta + \sin 2\theta \tan \alpha = 0$$

$$\sin 2\theta \tan \alpha = -\cos 2\theta$$

$$\tan 2\theta \tan \alpha = -1 \text{ ... as required}$$

iv. Show that $F'(\theta) = 0$, when $\theta = \frac{\alpha}{2} + \frac{\pi}{4}$

$$\text{If } \tan A \tan B = -1 \Rightarrow A - B = \frac{\pi}{2}$$

(Given in part (a)(ii) - see note in question)

Since $\tan 2\theta \tan \alpha = -1$ (see part (iii))

$$\Rightarrow 2\theta - \alpha = \frac{\pi}{2}$$

$$2\theta = \alpha + \frac{\pi}{2}$$

$$\theta = \frac{\alpha}{2} + \frac{\pi}{4}$$

Given that $F'(\theta) = 0$ when $\tan 2\theta \tan \alpha = -1$

$$\therefore F'(\theta) = 0 \text{ when } \theta = \frac{\alpha}{2} + \frac{\pi}{4} \text{ ... as required}$$

v. Since $v^2 = \frac{5d}{F(\theta)}$,

$$\Rightarrow v^2 \text{ is a MIN when } F(\theta) \text{ is a MAX}$$

We know $F'(\theta) = 0$ when $\theta = \frac{\alpha}{2} + \frac{\pi}{4}$

$$\therefore \text{MAX or MIN when } \theta = \frac{\alpha}{2} + \frac{\pi}{4}$$

$$F'(\theta) = \cos 2\theta + \sin 2\theta \tan \alpha$$

$$F''(\theta) = -2 \sin 2\theta + 2 \cos 2\theta \tan \alpha$$

$$\text{When } \theta = \frac{\alpha}{2} + \frac{\pi}{4}$$

$$F''\left(\frac{\alpha}{2} + \frac{\pi}{4}\right) = -2 \sin\left(\alpha + \frac{\pi}{2}\right) + 2 \cos\left(\alpha + \frac{\pi}{2}\right) \tan \alpha$$

Since $0 < \alpha < \frac{\pi}{2}$

$$\Rightarrow \tan \alpha > 0$$

$$\Rightarrow \sin\left(\alpha + \frac{\pi}{2}\right) > 0$$

$$\Rightarrow \cos\left(\alpha + \frac{\pi}{2}\right) < 0$$

$$\therefore F''\left(\frac{\alpha}{2} + \frac{\pi}{4}\right) < 0 \text{ i.e. MAX}$$

$\therefore v^2$ is a minimum.

♦♦ Mean mark part (iii) 21%
MARKER'S COMMENT: Many students failed to treat $\tan \alpha$ as a constant when differentiating.

♦♦♦ Mean mark 14%
MARKER'S COMMENT: Linking part (a)(ii) to this solution was a feature of the more efficient and successful approaches.

♦♦♦ Mean mark 9%
MARKER'S COMMENT: Only a small minority of students explained correctly that v^2 is a MIN when $F(\theta)$ is a MAX.

i. $x = 18t$

$$y = 18\sqrt{3}t - 5t^2$$

Particle hits slope when $y = -x$

$$18\sqrt{3}t - 5t^2 = -18t$$

$$5t^2 - 18t - 18\sqrt{3}t = 0$$

$$t(5t - 18 - 18\sqrt{3}) = 0$$

$$5t - 18 - 18\sqrt{3} = 0$$

$$5t = 18 + 18\sqrt{3}$$

$$t = \frac{18 + 18\sqrt{3}}{5}$$

$$\text{When } t = \frac{18 + 18\sqrt{3}}{5},$$

$$x = 18 \times \left(\frac{18 + 18\sqrt{3}}{5} \right)$$

Using Pythagoras (isosceles Δ):

$$\begin{aligned}
 OA &= \sqrt{2 \times 18^2 \times \left(\frac{18 + 18\sqrt{3}}{5} \right)^2} \\
 &= \sqrt{2} \times 18 \times \left(\frac{18 + 18\sqrt{3}}{5} \right) \\
 &= \frac{324(\sqrt{2} + \sqrt{6})}{5} \text{ units}
 \end{aligned}$$

ii. $x = 18t \Rightarrow \dot{x} = 18$

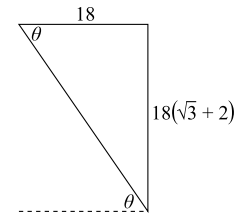
$$y = 18\sqrt{3}t - 5t^2 \Rightarrow \dot{y} = 18\sqrt{3} - 10t$$

$$\text{When } t = \frac{18 + 18\sqrt{3}}{5},$$

$$\begin{aligned}
 \dot{y} &= 18\sqrt{3} - 10 \left(\frac{18 + 18\sqrt{3}}{5} \right) \\
 &= 18\sqrt{3} - 36 - 36\sqrt{3} \\
 &= -18\sqrt{3} - 36
 \end{aligned}$$

$$= -18(\sqrt{3} + 2)$$

Find angle with the horizontal at impact:



$$\tan \theta = \frac{18(\sqrt{3} + 2)}{18}$$

$$= \sqrt{3} + 2$$

$$\theta = \tan^{-1}(\sqrt{3} + 2)$$

$$= 75^\circ$$

 $\therefore$ Angle made with the slope

$$= 75 - 45$$

$$= 30^\circ$$

♦ Mean mark part (ii) 39%.